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Date: ***13.03.2026 – 16.03.2026***
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BMW Global Sales Analysis (2018–2025)

Project Overview

The project analyzes BMW's global vehicle sales performance between **2018 and 2025**. The objective is to identify key drivers of sales and revenue growth, evaluate the impact of electric vehicle adoption, and assess the influence of macroeconomic factors. The analysis was conducted using **Microsoft Excel**, including **Power Query, Pivot Tables, and interactive dashboard.**

Business Question

What factors drive BMW's global sales and revenue performance across regions from 2018 to 2025?

Sub-questions

- Which **regions** generate the **most revenue**?
- Which **BMW models** contribute most to revenue?
- How has **EV adoption** evolved over time?
- Do **fuel prices** influence **EV adoption**?
- Does **GDP growth** influence **BMW sales**?

Dataset

Source of the dataset was obtained from Kaggle: **BMW Global Automotive Sales Dataset**

The dataset contains monthly BMW vehicle sales data with additional macroeconomic indicators.

Dataset Structure

The dataset contains **3,073 rows** and **11 columns**.

Column	Description
Year	Calendar year (2018–2025)
Month	Month of sale
Region	Market region (Europe, China, USA, Rest of World)
Model	BMW model
Units_Sold	Number of vehicles sold
Avg_Price_EUR	Average vehicle transaction price
Revenue_EUR	Total revenue
BEV_Share	Share of electric vehicles
Premium_Share	BMW premium market share
GDP[1]_Growth	Regional GDP growth (%)
Fuel_Price_Index	Relative fuel price index

[1] **Gross Domestic Product** is the total value of all goods and services produced in a country or region during a specific period (usually a year).

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Data Preparation

The dataset was imported into **Excel using Power Query**.

Steps performed:

1. Imported CSV dataset into Excel using **Power Query**.
2. Promoted headers to column names.
3. Adjusted column data types:
   - *Year* → Whole number
   - *Month* → Whole number
   - *Units_Sold* → Whole number
   - *Avg_Price_EUR* → Whole number
   - *Revenue_EUR* → Whole number
   - *BEV_Share* → Decimal number
   - *Premium_Share* → Decimal number
   - *GDP_Growth* → Decimal number
   - *Fuel_Price_Index* → Decimal number

Data Cleaning & Validation

- All columns were checked for missing values. **No missing values detected.**
- The dataset was checked for duplicate records based on the key fields: **Year**, **Month**, **Region**, and **Model**. **No duplicate rows were found**, confirming that each record represents a unique sales entry.
- Revenue values were verified using: **Revenue = Units_Sold × Avg_Price_EUR**

  The dataset showed **consistent** revenue calculations.

- During validation, **9 rows contained negative BEV share values**. Since EV market share cannot be negative, these rows were removed. This represented only **0.29% of the dataset**, indicating a negligible impact on the overall analysis.

Exploratory Data Analysis (EDA)

Several pivot analyses were created.

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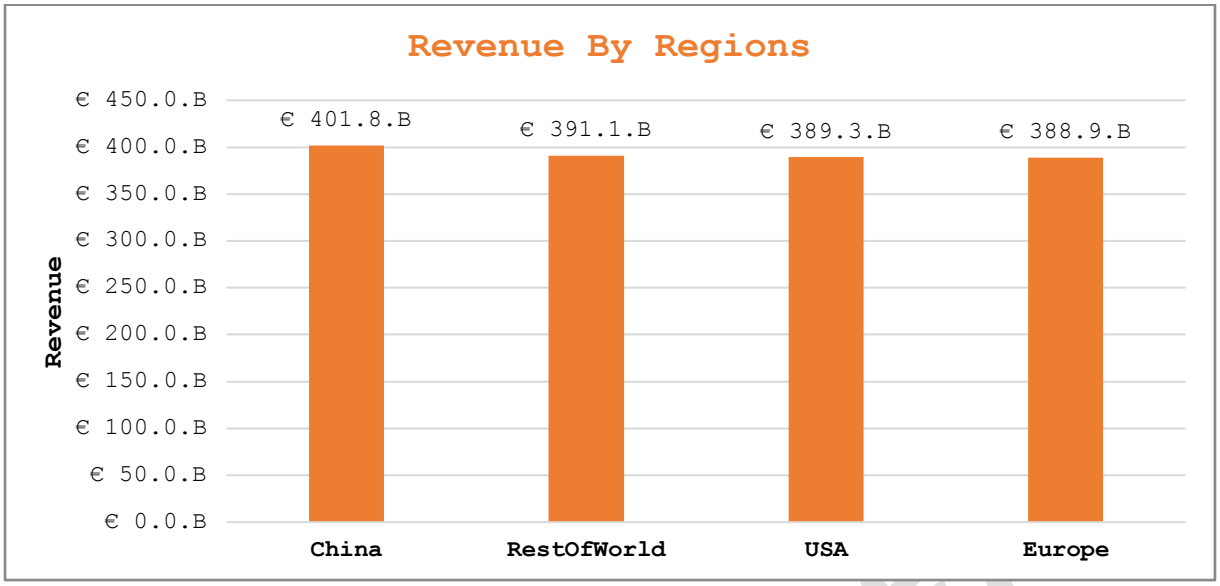
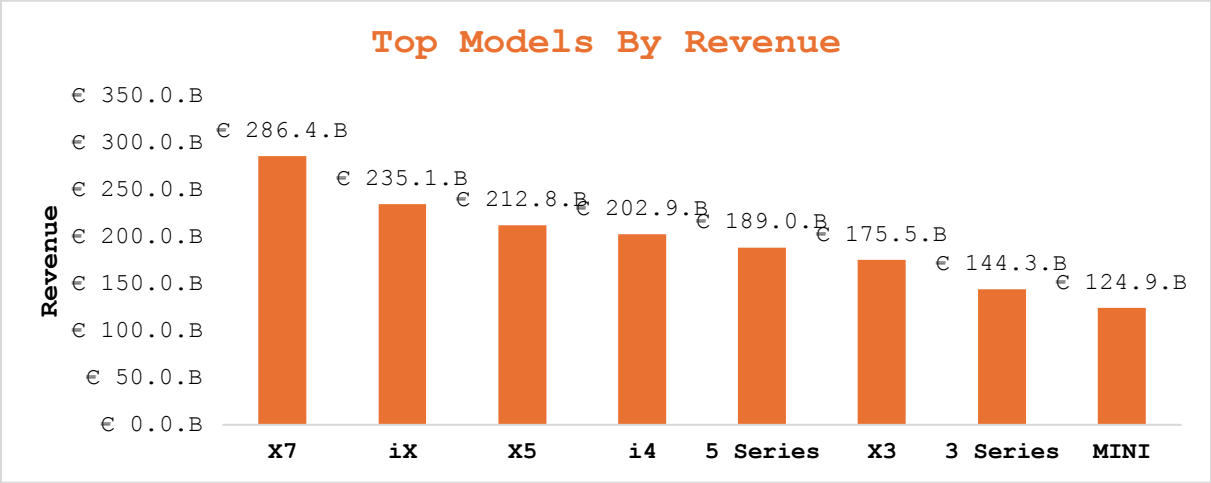
1. Regional Sales Performance



China generated the **highest BMW revenue** among all regions. This suggests that China is BMW's **largest and most valuable market**.

2. Model Revenue Performance

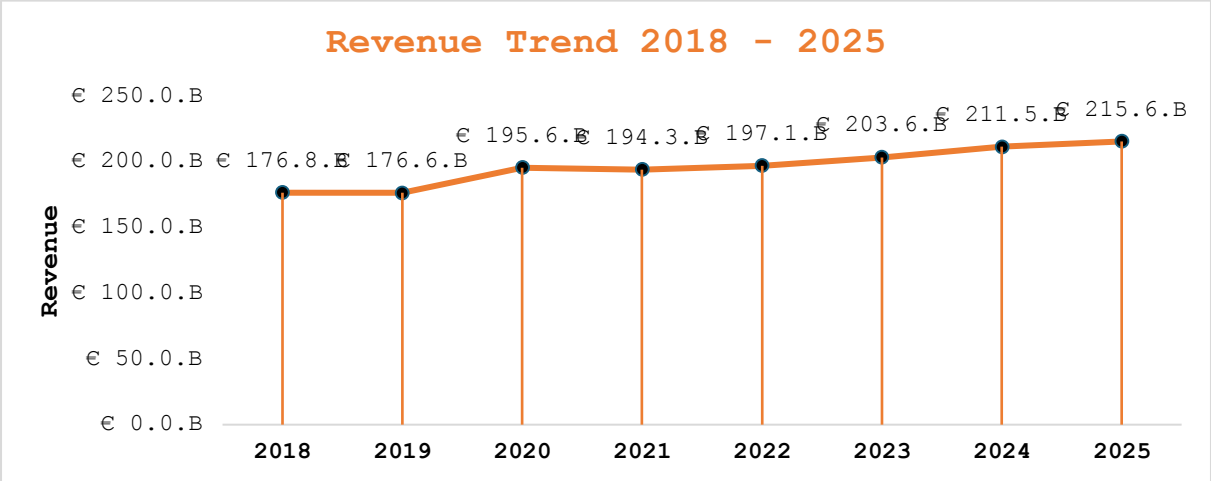


Top revenue-generating models:

1. **BMW X7**
2. **BMW iX**
3. **BMW X5**

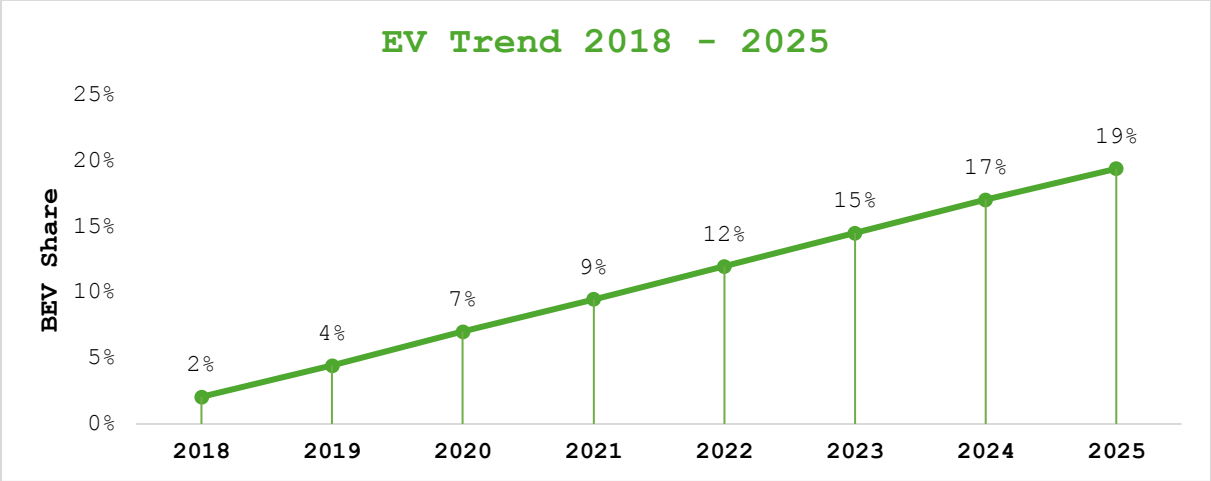
Premium SUV models contribute significantly to BMW revenue.

3. Revenue Trend



Revenue increased steadily from **2019 to 2025**, indicating strong business growth.

4. EV Adoption Trend



Electric vehicle share increased significantly. This reflects the rapid transition toward electric vehicles.

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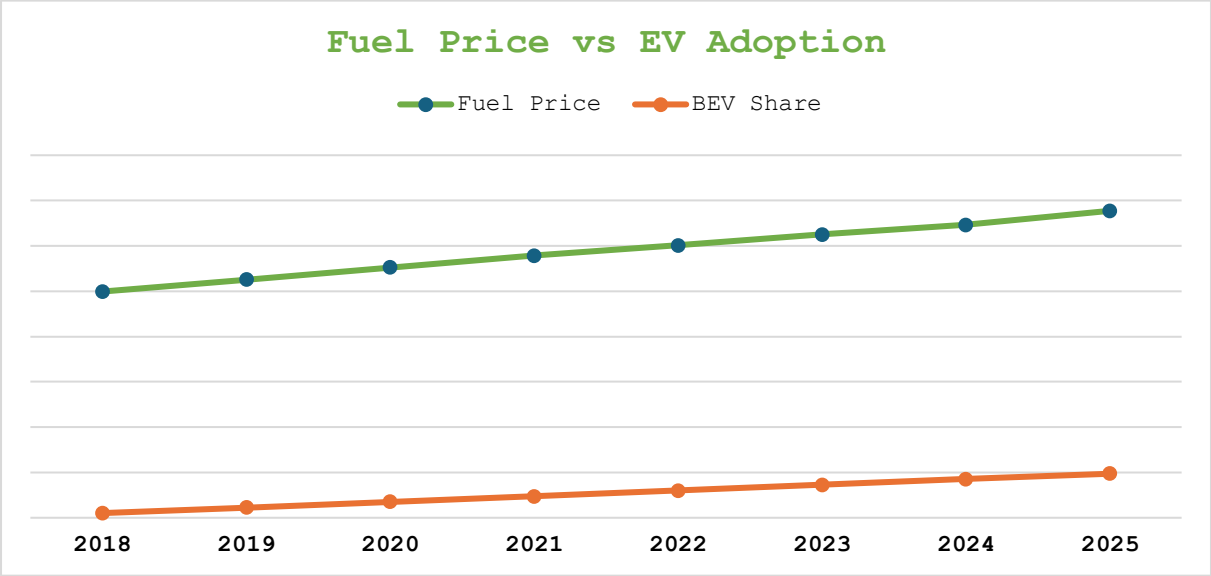
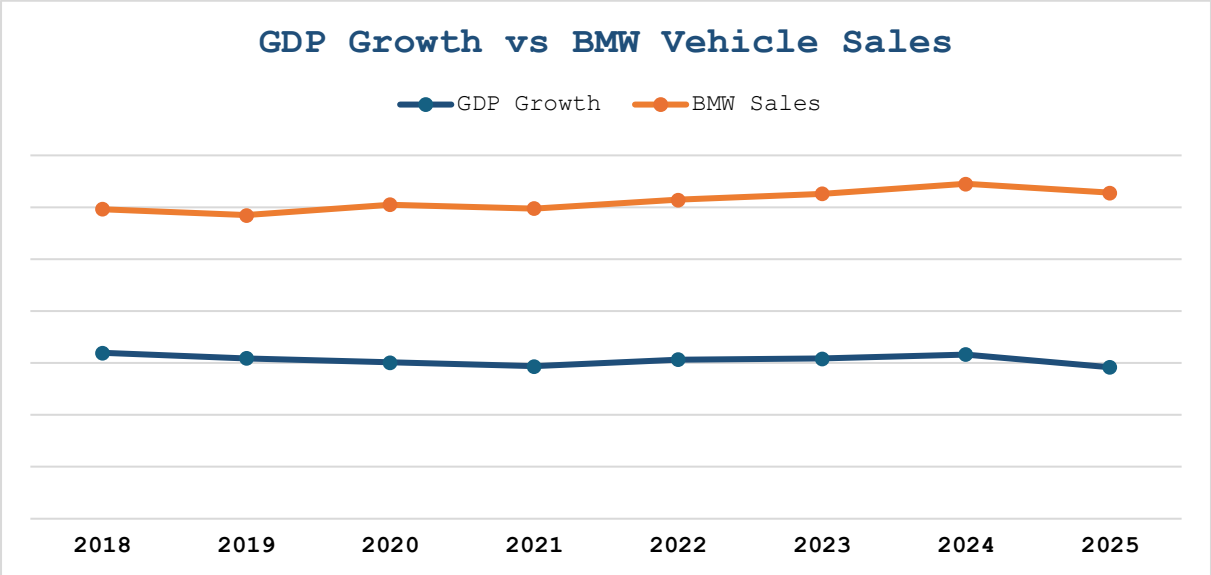
5. Fuel Price vs EV Adoption



Fuel price increases correlate with higher EV adoption. Evidently, **higher fuel costs may encourage consumers to switch to electric vehicles**.

6. GDP Growth vs Sales

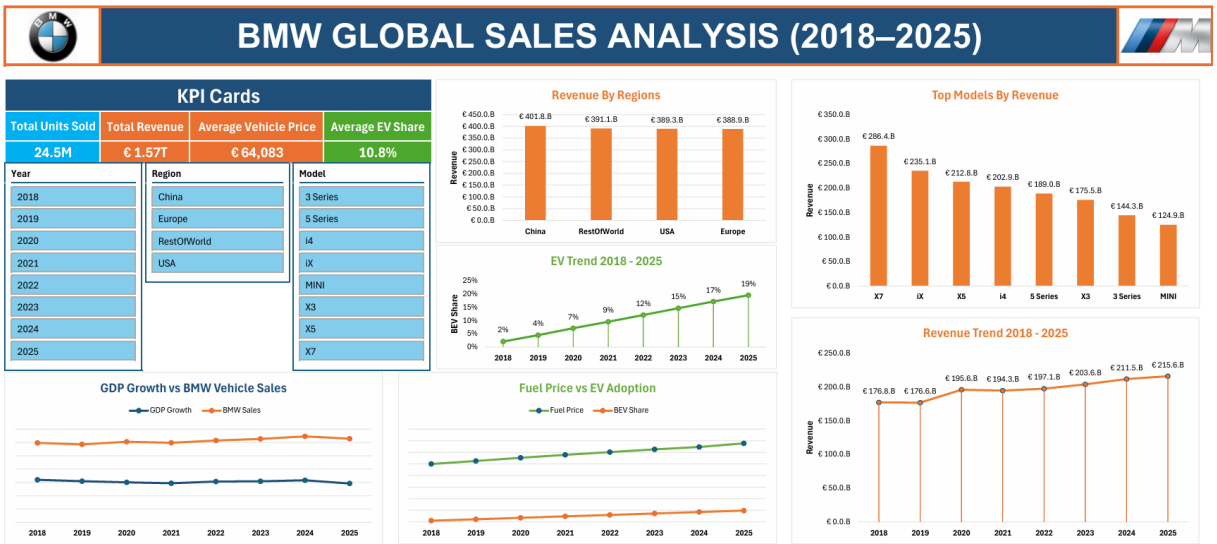


GDP growth showed **weak correlation with BMW sales** implying that other factors such as:

- product strategy
- regional demand
- EV expansion

may play a larger role in driving sales growth.

Dashboard



An **interactive Excel dashboard** was created to summarize insights.

Features include:

- KPI cards
- interactive slicers
- regional revenue analysis
- model revenue ranking
- EV adoption trends
- macroeconomic impact analysis

Dashboard KPIs:

- Total Units Sold: **24.5M**
- Total Revenue: **€1.57T**
- Average Vehicle Price: **€64,083**
- Average EV Share: **10.8%**

Key Insights

1. **China** is BMW's **largest revenue market**.
2. **Premium SUV** models drive the majority of BMW revenue.
3. Revenue has grown steadily **since 2019**.
4. **EV adoption** increased dramatically from **2%** to **19%**.
5. **Fuel prices** appear to influence **EV adoption**.
6. **GDP growth** has limited impact on **BMW sales performance**.

Tools Used

- Microsoft Excel
- Power Query
- Pivot Tables
- Pivot Charts
- Interactive Dashboard

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